

DATA STRUCTURE HOMEWORK 1

1.Sum of the two matrices

code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[10][10], b[10][10], sum[10][10];
```

```
    int r, c, i, j;
```

```
    printf("Enter number of rows and columns: ");
```

```
    scanf("%d %d", &r, &c);
```

```
    printf("Enter elements of Matrix A:\n");
```

```
    for(i = 0; i < r; i++)
```

```
        for(j = 0; j < c; j++)
```

```
            scanf("%d", &a[i][j]);
```

```
    printf("Enter elements of Matrix B:\n");
```

```
    for(i = 0; i < r; i++)
```

```
        for(j = 0; j < c; j++)
```

```
            scanf("%d", &b[i][j]);
```

```
    for(i = 0; i < r; i++)
```

```
        for(j = 0; j < c; j++)
```

```
            sum[i][j] = a[i][j] + b[i][j];
```

```
    printf("Sum of two matrices:\n");
```

```

for(i = 0; i < r; i++)
{
    for(j = 0; j < c; j++)
        printf("%d ", sum[i][j]);
    printf("\n");
}

return 0;
}

```

Output:

Enter number of rows and columns: 2 2

Enter elements of Matrix A:

2 3

3 5

Enter elements of Matrix B:

3 6

5 8

Sum of two matrices:

5 9

8 13

2. Read n values and display in reverse order.

code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```

int a[100], n, i;

printf("Enter n: ");
scanf("%d", &n);

printf("Enter array elements:\n");
for(i = 0; i < n; i++)
    scanf("%d", &a[i]);

printf("Reverse order:\n");
for(i = n - 1; i >= 0; i--)
    printf("%d ", a[i]);

return 0;
}

```

output:

Enter n: 5

Enter array elements:

20 40 50 80 10

Reverse order:

10 80 50 40 20

3.count total duplicate elements in an array.

code:

```
#include <stdio.h>
```

```
int main()
```

```

{
    int a[100], n, i, j, count = 0;

    printf("Enter n: ");
    scanf("%d", &n);

    printf("Enter array elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for(i = 0; i < n; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(a[i] == a[j])
            {
                count++;
                break;
            }
        }
    }

    printf("Total duplicate elements = %d", count);

    return 0;
}

```

output:

Enter n: 6

Enter array elements:

30 40 50 30 40 60

Total duplicate elements = 2

4.print unique element in an array.

code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[100], n, i, j, count;
```

```
    printf("Enter n: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter elements:\n");
```

```
    for(i = 0; i < n; i++)
```

```
        scanf("%d", &a[i]);
```

```
    printf("Unique elements: ");
```

```
    for(i = 0; i < n; i++)
```

```
    {
```

```
        count = 0;
```

```
        for(j = 0; j < n; j++)
```

```
        {
```

```
            if(a[i] == a[j])
```

```

        count++;
    }

    if(count == 1)
        printf("%d ", a[i]);
    }

    return 0;
}

```

output:

Enter n: 6

Enter elements:

30 60 70 50 50

60

Unique elements: 30 70

5. Insert an element in a sorted array.

code: #include <stdio.h>

```

int main()
{
    int a[100], n, value, pos, i;

    printf("Enter n: ");
    scanf("%d", &n);

    printf("Enter sorted elements:\n");

```

```
for(i = 0; i < n; i++)  
    scanf("%d", &a[i]);  
  
printf("Enter element to insert: ");  
scanf("%d", &value);  
  
pos = n;  
  
for(i = 0; i < n; i++)  
{  
    if(value < a[i])  
    {  
        pos = i;  
        break;  
    }  
}  
  
for(i = n; i > pos; i--)  
    a[i] = a[i - 1];  
  
a[pos] = value;  
n++;  
  
printf("Array after insertion:\n");  
for(i = 0; i < n; i++)  
    printf("%d ", a[i]);  
  
return 0;
```

```
}
```

output:

Enter n: 6

Enter sorted elements:

50

60

80

40

55

36

Enter element to insert: 50

Array after insertion:

50 50 60 80 40 55 36

6.Insert an element in an unsorted array.

code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[100], n, value, pos, i;
```

```
    printf("Enter n: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter array elements:\n");
```

```
    for(i = 0; i < n; i++)
```

```
scanf("%d", &a[i]);

printf("Enter position: ");
scanf("%d", &pos);

printf("Enter element: ");
scanf("%d", &value);

for(i = n; i >= pos; i--)
    a[i] = a[i - 1];

a[pos - 1] = value;
n++;

printf("Array after insertion:\n");
for(i = 0; i < n; i++)
    printf("%d ", a[i]);

return 0;
}
```

output:

Enter n: 6

Enter array elements:

50

80

70

40

33

58

Enter position: 5

Enter element: 60

Array after insertion:

50 80 70 40 60 33 58

7. Multiplication of matrix.

code

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[10][10], b[10][10], c[10][10];
```

```
    int r1, c1, r2, c2, i, j, k;
```

```
    printf("Enter rows and columns of Matrix A: ");
```

```
    scanf("%d %d", &r1, &c1);
```

```
    printf("Enter rows and columns of Matrix B: ");
```

```
    scanf("%d %d", &r2, &c2);
```

```
    if(c1 != r2)
```

```
    {
```

```
        printf("Matrix multiplication not possible.");
```

```
        return 0;
```

```
    }
```

```
printf("Enter Matrix A:\n");  
for(i = 0; i < r1; i++)  
    for(j = 0; j < c1; j++)  
        scanf("%d", &a[i][j]);  
  
printf("Enter Matrix B:\n");  
for(i = 0; i < r2; i++)  
    for(j = 0; j < c2; j++)  
        scanf("%d", &b[i][j]);  
  
for(i = 0; i < r1; i++)  
    for(j = 0; j < c2; j++)  
    {  
        c[i][j] = 0;  
        for(k = 0; k < c1; k++)  
            c[i][j] += a[i][k] * b[k][j];  
    }  
  
printf("Result:\n");  
for(i = 0; i < r1; i++)  
{  
    for(j = 0; j < c2; j++)  
        printf("%d ", c[i][j]);  
    printf("\n");  
}  
  
return 0;
```

```
}
```

output

Enter rows and columns of Matrix A: 2

2

Enter rows and columns of Matrix B: 2

2

Enter Matrix A:

2 3

4 5

Enter Matrix B:

5 6

7 8

Result:

31 36

55 64

8.Matrix transpose.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[10][10], r, c, i, j;
```

```
    printf("Enter rows and columns: ");
```

```

scanf("%d %d", &r, &c);

printf("Enter matrix:\n");
for(i = 0; i < r; i++)
    for(j = 0; j < c; j++)
        scanf("%d", &a[i][j]);

printf("Transpose:\n");
for(j = 0; j < c; j++)
{
    for(i = 0; i < r; i++)
        printf("%d ", a[i][j]);
    printf("\n");
}

return 0;
}

```

output:

Enter rows and columns: 2 3

Enter matrix:

2 5 6

6 8 9

Transpose:

2 6

5 8

6 9

9. find the right diagonal sum.

code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[10][10], n, i, j, sum = 0;
```

```
    printf("Enter order of matrix: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter matrix:\n");
```

```
    for(i = 0; i < n; i++)
```

```
        for(j = 0; j < n; j++)
```

```
            scanf("%d", &a[i][j]);
```

```
    for(i = 0; i < n; i++)
```

```
        sum += a[i][n - 1 - i];
```

```
    printf("Right diagonal sum = %d", sum);
```

```
    return 0;
```

```
}
```

output:

Enter order of matrix: 2

Enter matrix:

2 4

5 6

Right diagonal sum = 9

10. check whether a matrix is an identity matrix.

code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[10][10], n, i, j, flag = 1;
```

```
    printf("Enter order of matrix: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter matrix:\n");
```

```
    for(i = 0; i < n; i++)
```

```
        for(j = 0; j < n; j++)
```

```
            scanf("%d", &a[i][j]);
```

```
    for(i = 0; i < n; i++)
```

```
    {
```

```
        for(j = 0; j < n; j++)
```

```

{
    if(i == j && a[i][j] != 1)
        flag = 0;

    if(i != j && a[i][j] != 0)
        flag = 0;
}
}

if(flag)
    printf("Identity Matrix");
else
    printf("Not an Identity Matrix");

return 0;
}

```

output:

Enter order of matrix: 2

Enter matrix:

1 0

0 1

Identity Matrix

11.count vowels,constants,digits and white spaces.

code:

```
#include <stdio.h>
```

```
#include <ctype.h>

int main()
{
    char str[200];

    int i, vowels = 0, consonants = 0;

    int digits = 0, spaces = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(isalpha(str[i]))
        {
            if(str[i]=='a' || str[i]=='e' || str[i]=='i' ||
               str[i]=='o' || str[i]=='u' ||
               str[i]=='A' || str[i]=='E' || str[i]=='I' ||
               str[i]=='O' || str[i]=='U')
                vowels++;
            else
                consonants++;
        }
        else if(isdigit(str[i]))
            digits++;
        else if(isspace(str[i]))
            spaces++;
    }
}
```

```
printf("Vowels = %d\n", vowels);  
printf("Consonants = %d\n", consonants);  
printf("Digits = %d\n", digits);  
printf("White spaces = %d\n", spaces);  
  
return 0;  
}
```

output:

Enter a string: monika@2008

Vowels = 3

Consonants = 3

Digits = 4

White spaces = 1

12.find frequency of a character in a string.

code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char str[200], ch;
```

```
    int i, count = 0;
```

```
    printf("Enter a string: ");
```

```
    fgets(str, sizeof(str), stdin);
```

```

printf("Enter character: ");

scanf("%c", &ch);


for(i = 0; str[i] != '\0'; i++)
{
    if(str[i] == ch)
        count++;
}


printf("Frequency of '%c' = %d", ch, count);


return 0;
}

```

output:

Enter a string: monika

Enter character: o

Frequency of 'o' = 1

13.count number of words in a string.

code:

```
#include <stdio.h>
```

```
#include <ctype.h>
```

```
int main()
```

```
{  
    char str[200];  
    int i, words = 0, inWord = 0;  
  
    printf("Enter a string: ");  
    fgets(str, sizeof(str), stdin);  
  
    for(i = 0; str[i] != '\0'; i++)  
    {  
        if(!isspace(str[i]) && inWord == 0)  
        {  
            words++;  
            inWord = 1;  
        }  
        else if(isspace(str[i]))  
        {  
            inWord = 0;  
        }  
    }  
  
    printf("Number of words = %d", words);  
  
    return 0;  
}
```

output:

Enter a string: monika

Number of words = 1

14. sort string array.

code

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char str[100][50], temp[50];
```

```
    int n, i, j;
```

```
    printf("Enter number of strings: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter strings:\n");
```

```
    for(i = 0; i < n; i++)
```

```
        scanf("%s", str[i]);
```

```
    for(i = 0; i < n - 1; i++)
```

```
    {
```

```
        for(j = i + 1; j < n; j++)
```

```
        {
```

```
            if(strcmp(str[i], str[j]) > 0)
```

```
            {
```

```
                strcpy(temp, str[i]);
```

```
                strcpy(str[i], str[j]);
```

```
                strcpy(str[j], temp);
```

```
            }
```

```
    }  
}  
  
printf("Sorted strings:\n");  
for(i = 0; i < n; i++)  
    printf("%s\n", str[i]);  
  
return 0;  
}
```

output:

Enter number of strings: 5

Enter strings:

susi

moni

abi

sindhu

harini

Sorted strings:

abi

harini

moni

sindhu

susi

15.toggle case of a sentence

code:

```
#include <stdio.h>
```

```
#include <ctype.h>

int main()
{
    char str[200];
    int i;

    printf("Enter a sentence: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(islower(str[i]))
            str[i] = toupper(str[i]);
        else if(isupper(str[i]))
            str[i] = tolower(str[i]);
    }

    printf("Toggled sentence:\n%s", str);

    return 0;
}
```

output

Enter a sentence: i am monika

Toggled sentence:

I AM MONIKA

